Capstone Design Notebook

Car Rider Pickup System

CSCE 483

Instructor: Rabi Mahapatra
Sara Vorhaben

TEXAS A&M UNIVERSITY
Department of Computer
Science & Engineering

Week 1 (January 12 - January 16):

Entry: #1

Class Journal Entry: January 12, 2026

Teammates Present: All potential teammates as teams haven't been decided but Jack Uehlinger, Jaden Pearson, Haya Absuada, and Jake Hewitt were all there.

What was accomplished in class: Today we talked about potential groupmates, the scope of the course & its requirements, and potential project ideas

Action items before next class

1. Order Textbook
2. Think about project ideas
3. Turn in questionnaire (https://docs.google.com/document/d/1LYhKTf_DagKlgS9f1H3CBgjT-Gc_9nhG/edit)

Entry: #2

Personal Work Journal Entry: January 14, 2026

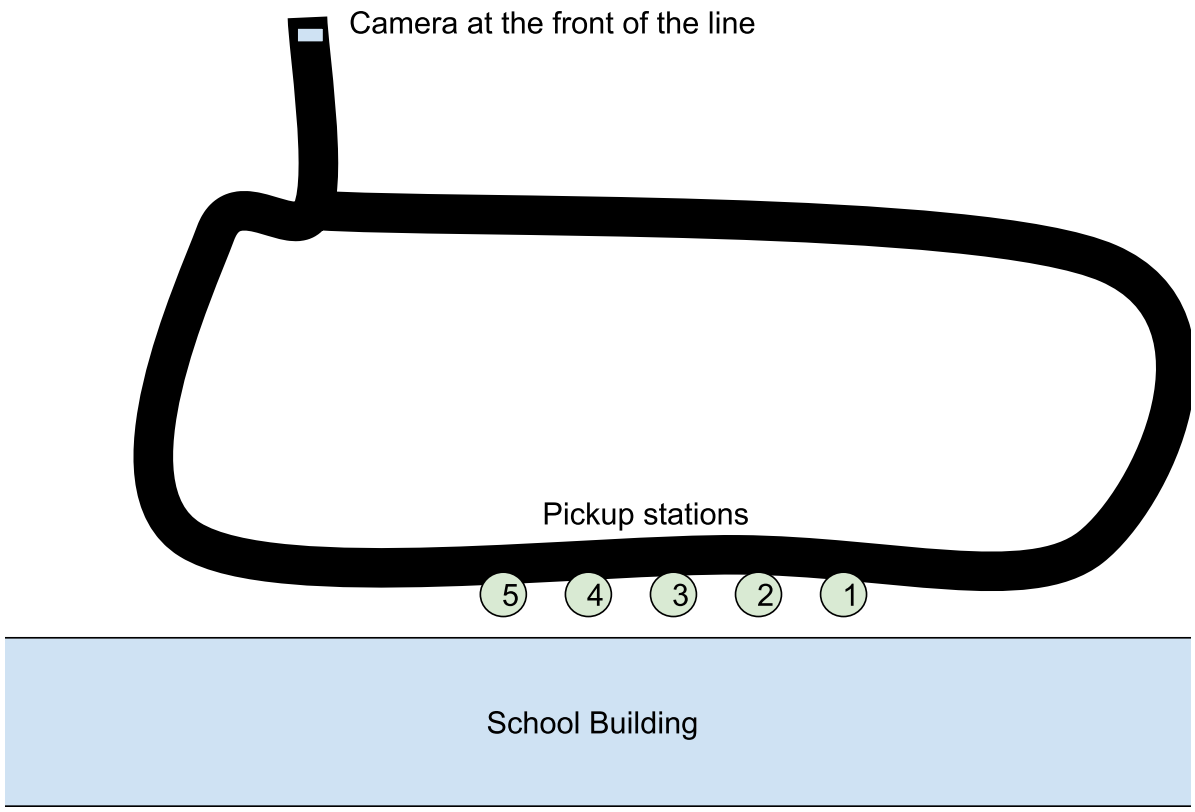
Teammates Present (if any): None

Updates on Previous Action Items:

1. Turned in questionnaire

Potential Problem to solve:
We want to make school dropoff more streamlined to decrease the time needed to pick up elementary students from school

My Proposed solution:
Use a camera to read the license plates of the cars in the order that they drive into the pickup line. Using the licence plates of the parents, match that to students (in a database) and then send this information to the people in the front of the line(in the school) and inform them that their kid is ready to be picked up and which station to go to(see diagram below):



Things needed to make this successful:

- Camera that reads license plates
    - Idk much about this need to look into it more
- System that matches license plates to the correct kid (securely)
    - Create a site where parents can enter their license plate number and kids information(verified by the school)
    - Once info is entered into the site, query the database based on the license plate and determine the correct kid
- A way to determine which station they will be at when you get to the front of the line
    - Keep track of the number of cars in/out and determine which station the car will go to
- A way to get the information to the teachers/admin to tell the students where to go
    - Send the info via (Wifi? Bluetooth?) to a laptop or other device at the front of the school

<u>Questions for Teammates/Advisors</u>:

Entry: #3

Class Journal Entry: January 14, 2026

Teammates Present: All Potential teammates present

Updates on Previous Action Items: [No previous Action Items]

What was accomplished in class: In class today we finalized and submitted our potential team (the Wranglers) and learned about teamwork. Additionally, we continued to talk about project ideas. I think we have decided to go with the School Drop off idea. We began to discuss how we could go about designing and implementing it so it could work. Additionally, I created a google drive to house all of our ideas so that everyone had access to it.

Action items before next class:
1. Research in depth how to use facial recognition and how to connect an arduino to a website (and keep it deployed).

Questions for Teammates/Advisors:
Would WIFI/Bluetooth be better for connection between the camera in the parking lot and the laptop running the program at the front of the school?

Entry: #4

Personal Work Journal Entry: January 15, 2026

Teammates Present (if any): none

Updates on Previous Action Items:
1. Research in depth how to use facial recognition and how to connect an arduino to a website (and keep it deployed).:
Probably best not to use an arduino. Pi would be much easier and can do more + implement camera better

Week 2

Entry: #5

Personal Work Journal Entry: January 20, 2026

Updates on Previous Action Items:
1. 1. Research in depth how to use facial recognition and how to connect an arduino to a website (and keep it deployed).:

How to connect a Raspberry Pi to a website (not too difficult):
Recommended arch:

- React frontend
- Node backend
- Server communication
- Raspberry pi
- Camera access (raspberry pi)

Src:
https://vsik.hashnode.dev/serving-a-react-app-from-a-raspberry-pi
https://www.instructables.com/Stream-Raspberry-Pi-Camera-and-Microphone-to-React/

How to use facial recognition
https://realpython.com/face-recognition-with-python/

2. I worked on establishing necessary fields for the database to ensure that we weren't missing anything extreme and created a dbdiagram for potential tables
   https://dbdiagram.io/d/Wranglers-6967ef4bd6e030a024100713
   Going to the team to look at it and give feedback

Questions/Comments from team members/advisors:

1. Do we need more tables in the database (student bc can pick up an array of students)?

Entry: #6

Class Journal Entry: January 21, 2026

Teammates Present: All Teammates present

What was accomplished in class:

- Today we talked about project ideas with the professor.
- We decided on our project to be a school drop off system.
- Today we started coming up with potential solutions. We wrote a list of pros and cons for several different systems.
    - I came up with a few options (licence plate reader and pin input) and we debated the pros and cons of the options.
- Went over the db diagram I created and made changes - nowhere near final but is at least a starting point

Action items before next class:
1. Research more in depth the wifi/bluetooth issues with the pi being so far from the system
2. Licence plate reader how to do

Questions for Teammates/Advisors:
What is the school drop off layout?
What do they have to consider for custody of children?
What information do they currently store for students? (does this include licence plate numbers and drivers licences)

Week 3

Entry: #7

Class Journal Entry: January 28, 2026

Teammates Present: Me, Jaden, Jack, Jake (Hiya was missing today, she went to the career fair)

What was accomplished in class:
Monday was a snow day so no entry from then. What I did today:

- Lecture on forming problem statement
- Learned about needs statement and how not to include how in the needs statement
- Talked with the professor about getting a meeting set up for next week with the school
- I worked on writing the needs statement
- I worked on writing the objective statement
- We began discussing questions to ask the school next week to ensure that we have all information we need to be successful
- I drew out a model of how the pickup stations would look based on the school

Action items before next class:
1. Research more in depth on how the school pickup works
2. Come up with potential interface ideas
3. Add more detail to the objective/needs statements

Questions for Teammates/Advisors:

Week 4:

Entry: #8

Class Journal Entry: February 02, 2026

Teammates Present: All teammates were present today

Updates on Previous Action Items:
1. Research more in depth on how the school pickup works: Still looking into different possibilities - right now the leading one is the licence plate reader but we could potentially do qr codes or facial rec depending on more data
2. Come up with potential interface ideas: looked into this a little but still need to do more research
3. Add more detail to the objective/needs statements: Created a doc with the potential needs statements

What was accomplished in class:
- We had a lecture about designing the requirements for our project
- We talked about how we would need a big battery in order to use the radio connections- need to think about other options
- We finalized our Needs statement and Objective statement
- We began working on requirements for the design

Action items before next class:
1. Create a logo for our team (:
2. Look into website design information/etc
3. Work on finalizing the requirements for the project/formatting

Project Status:
Behind **On Time** Ahead
We are moving at a decent pace and have constructed needs statements, however we still need to meet with the client.

Questions for Teammates/Advisors:
- When is our meeting with the school?
- Are there templates for the Objective/Needs statement we need to use

Entry: #9

Personal Work Journal Entry: February 04, 2026

Teammates Present (if any): None Present

Updates on Previous Action Items:
1. Create a logo for our team (:
I worked on the logo for the Team, I drew it in notability and then uploaded it to canva, going to get team feedback and see what needs to be changed. My design for the team logo can be found here: https://www.canva.com/design/DAHAYJvwVV0/9ylcvPqPrSsB6uzBOC5CYw/edit

2. Look into website design information/etc:
Began looking into what the site design will look like. We for sure need a parent portal/profile page, home screen that displays who is being picked up (names etc.) and a school profile page for updates.
Parent Profile page:
- Displays kids names
- Under each kid it lists who is eligible to pick them up (and when)
- Has an option to update information (sends request to school to say when they will be picked up)
- Shows pending requests and last updated date

Home Screen:
- Shows stations 1-6 and order of kids (ex station1:Suzie, station2: john)

School Admin Screen:
- Lists all names and registered parents (table format)
- Has a requests page that includes all requests for updates to information (admin can click on each request, read it and confirm it so it reflects in the system)
- Has a temporary change page for if someone is switching to be a car rider today

Going to get team feedback and see what I should change. If all looks good I'll probably do a figma in the next couple of days

3. Work on finalizing the requirements for the project/formatting:
I began working on the template for the slides, going to see if people like it and keep going

Questions/Comments from team members/advisors:
- Can you look at slides
- Get Prof to look at need/objective statements

Entry: #10

Class Journal Entry: February 04, 2026

Teammates Present:

Updates on Previous Action Items:
I worked on the action items (Create a logo for our team, Look into website design information/etc, and Work on finalizing the requirements for the project/formatting) in my personal work entry above

What was accomplished in class:

- Talked about our needs and objective statements as a team
- Got feedback from the professor and updated our needs and objective statements
- Submitted our needs and objective statements
- Split up proposal report so we could begin to work on it
- Continued to research the background/other information to build the best program
- Continued to research our wifi connection dilemma
-

Action items before next class:
1. Work on gantt chart
2. Create a slideshow presentation and add to the team drive
3. Begin thinking about figma/design elements for website

Questions for Teammates/Advisors:
Engineering Design: What would be the best wifi/bluetooth connection?
Could the prof look over the requirements for us?
Are we having a meeting? If so, when? Are we doing it specifically for this school?

Week 5:

Entry: #11

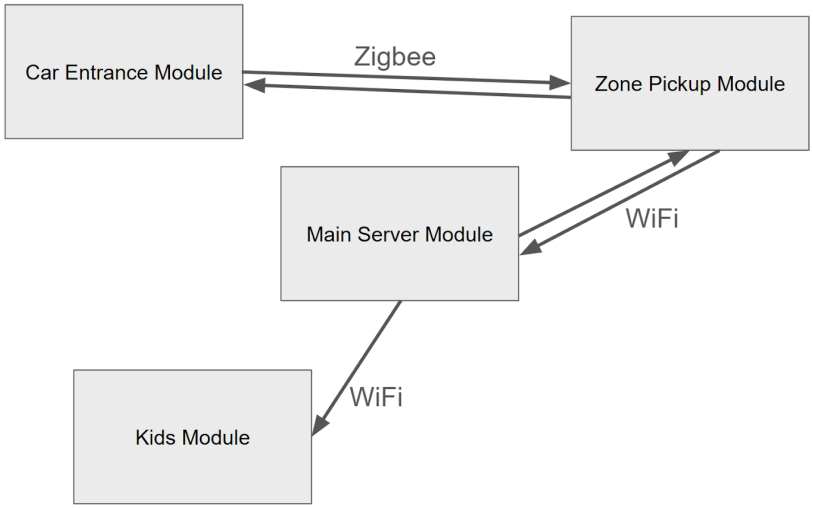
Class Journal Entry: February 09, 2026

Teammates Present: Everyone present and early because of the quiz

Updates on Previous Action Items:
1. Work on gantt chart: monday -> progress but not done
2. Create a slideshow presentation and add to the team drive: slideshow added
3. Begin thinking about figma/design elements for website: not started yet

What was accomplished in class:

- Took the Five Dysfunctions of a team quiz
- Met with the prof and talked about team requirements
- Talked through project setup



- Delegated technical responsibilities
- Discussed presentation
- Created [Monday Task management]
- Jack went with the TA to get PIs, Cameras and Screens 🙂
- Turned in the requirement document

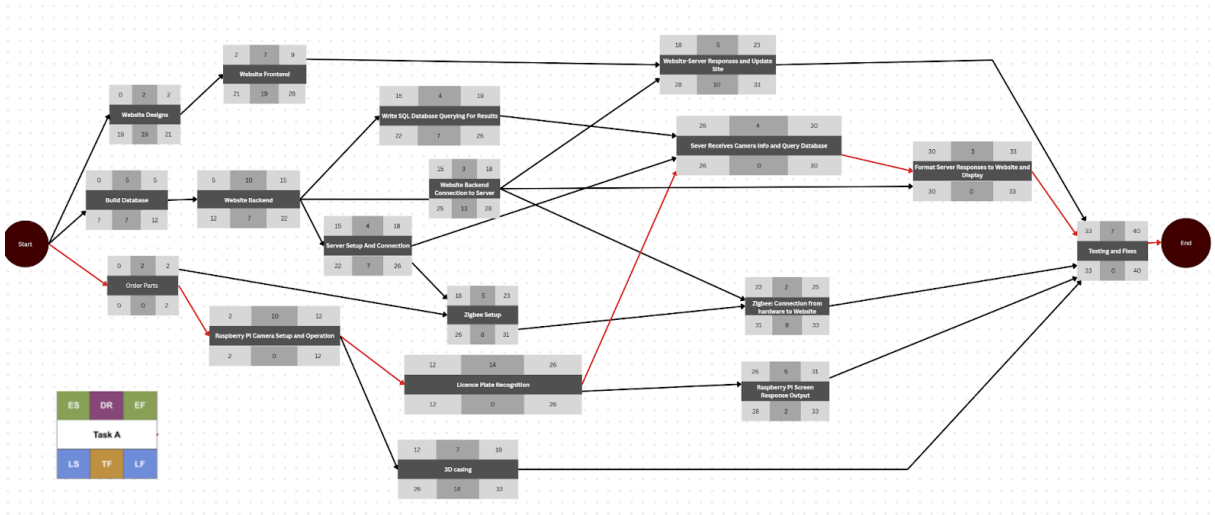
Action items before next class:
1. Get Slides done by tomorrow
2. Finish Monday (task chart)
3. Finish other diagram + PERT chart

Entry: #12

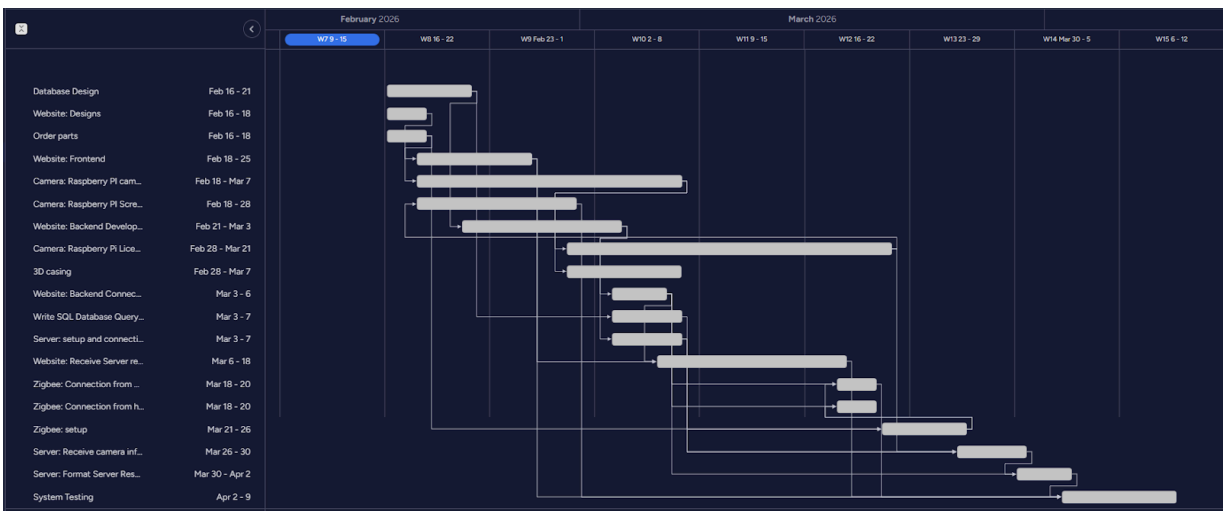
Personal Work Journal Entry: February 10, 2026

Teammates Present (if any): None Present

Updates on Previous Action Items:
1. 1. Get Slides done by tomorrow:
Working on slides:
Schedule of tasks, Pert and Gantt charts (1 page, 5 pts)
Built pert chart and Gantt chart:
PERT CHART: Link to view pert chart online (more readable



GAUNT CHART: I added all of the events into our [Monday Project Manager]() and created a gaunt chart seen below:



I also worked on economic analysis and social safety and environment analysis slides from our [presentation.]()

Entry: #13

Class Journal Entry: February 11, 2026

Teammates Present: Everyone was there

Updates on Previous Action Items:

1. Get Slides done by tomorrow: finished
2. Finish Monday (task chart): finished last night
3. Finish other diagram + PERT chart: completed

What was accomplished in class:

- Went over presentations -> We presented our slides to the class and listened to the other presentations
- Talked over our slides after presentation and got feedback so we can ensure our proposal is correct
- Checked in with groupmates about proposal progress: what still needs to be done and what can be improved, everyone seems to be doing fine and is almost done

Action items before next class:
1. Complete Project proposal
2. Begin thinking about database design/ initial setup + Figma frontend designs
3. Prof mentioned license plate might be hard to make accurate, look into alternatives/other options

Questions for Teammates/Advisors:
Na

Week 6:

Entry: #14

Class Journal Entry: February 16, 2026

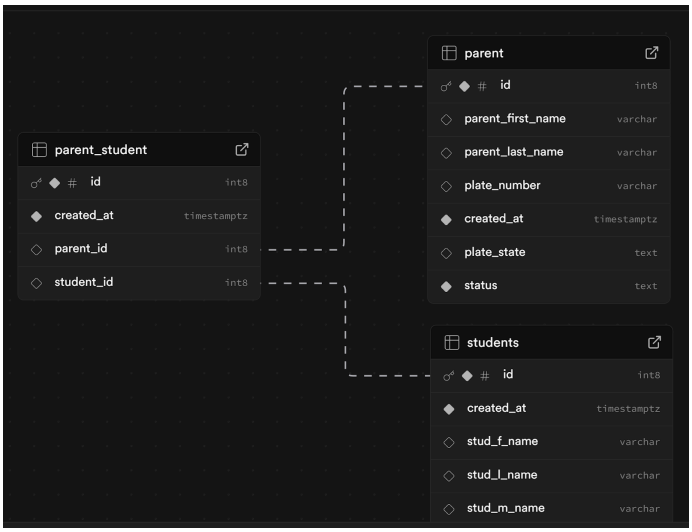
Teammates Present: All teammates present

Updates on Previous Action Items:
1. Complete Project proposal: completed
2. Begin thinking about database design/ initial setup + Figma frontend designs: started
3. Prof mentioned license plate might be hard to make accurate, look into alternatives/other options: looked into qr codes more

What was accomplished in class:

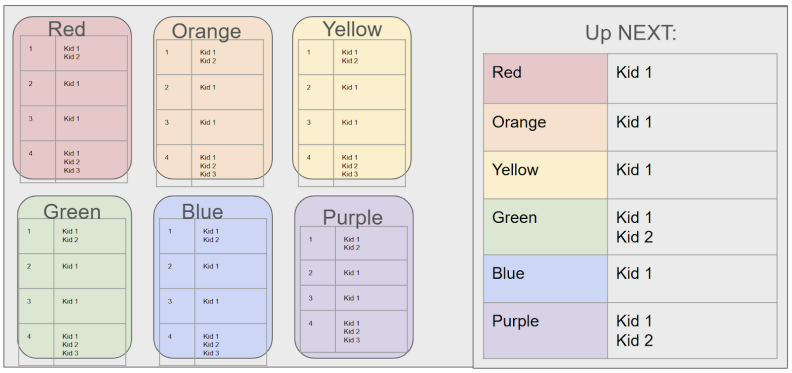
- Met with the prof and went over proposal presentation slides again.
    - Got more feedback and are going to reduce hardware on zone pickup module
    - May implement phones at pickup stations for station handlers

- Set up the database in Supabase



- Created a parts list to order parts- > Hiya will try and order them by wednesday
- Created wireframes for our website: https://docs.google.com/presentation/d/1N-_XUV2rMSEdKnJUqgwT66DcYqOZtxBdcBhvxXKX5aE/edit?slide=id.g3c8750d00b9_0_11#slide=id.g3c8750d00b9_0_11

Kids Module Screen

- 

Action items before next class:

1. Write RLS for database
2. Join the github

Project Status:

Behind **On Time** Ahead

We are moving at the expected pace.

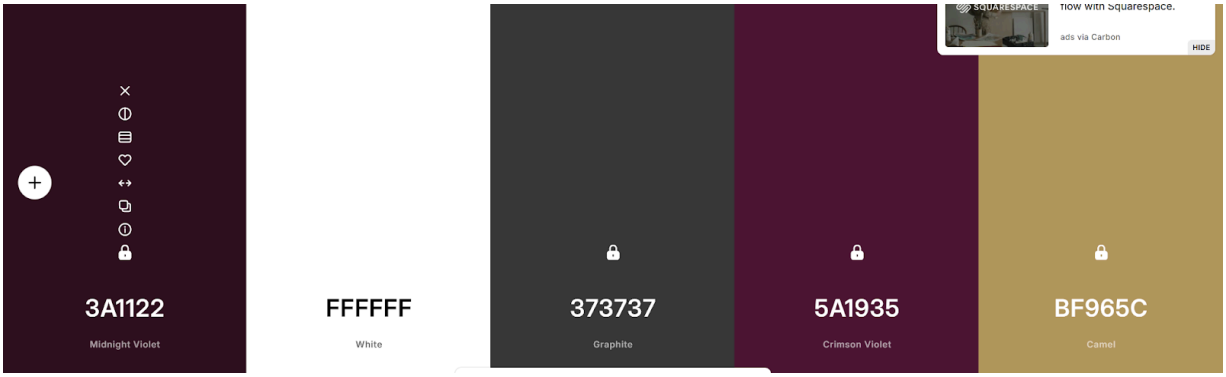
Entry: #15

Class Journal Entry: February 18, 2026

Teammates Present: All members present

Updates on Previous Action Items:
1. Write RLS for database: still working on it
2. Join the github: joined the github and cloned the repo

What was accomplished in class:
- All of the wireframes were completed and I added descriptions talking about what each would do
- Hiya added a readme to the repo
- Jack, Jaden and jake worked on the raspberry pi -> it is not being cooperative
- I added google oath to the database
- Jayden created an Onshape (to be used for the housing of the pi and camera and shared it with us and I joined
- Picked a website theme



Action items before next class:
1. Keep working on RLS Policies
2. React Project

Project Status:
Behind **On Time** Ahead

Questions for Teammates/Advisors:
RAHHHH

Week 7:

Entry: #16

Class Journal Entry: February 23, 2026

Teammates Present: All Teammates Present

Updates on Previous Action Items:
1. Keep working on RLS Policies: in progress
2. React Project: react project created and pages are beginning to be implemented

What was accomplished in class: Kept working on the Parent Portal Page to ensure all information was able to be accessed in a secure manner. I also implemented an update page for if parents need to update their info. Additionally we talked with the prof for our weekly meeting so that we were up to date on everything.

Action items before next class:
1. Connect Database
2. Complete Parent Portal page
3. Look into QR APIs to implement

Project Status:
Behind **On Time** Ahead

Questions for Teammates/Advisors:

Entry: #17

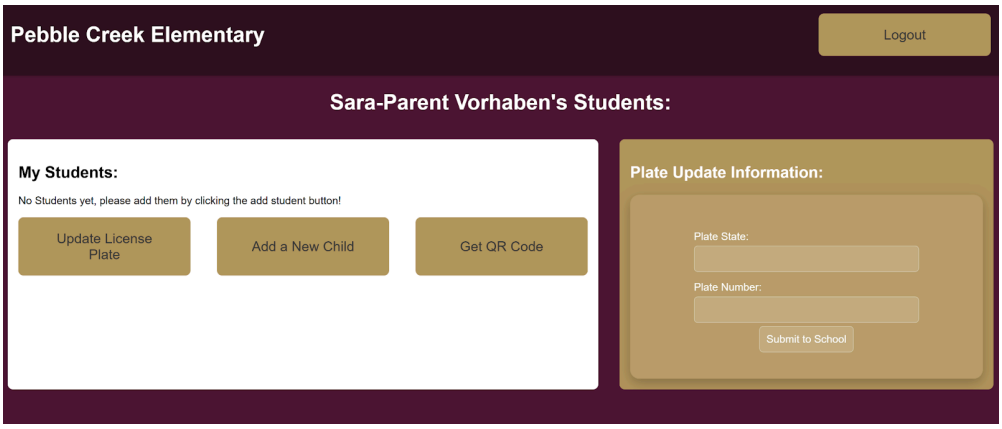
Class Journal Entry: February 25, 2026

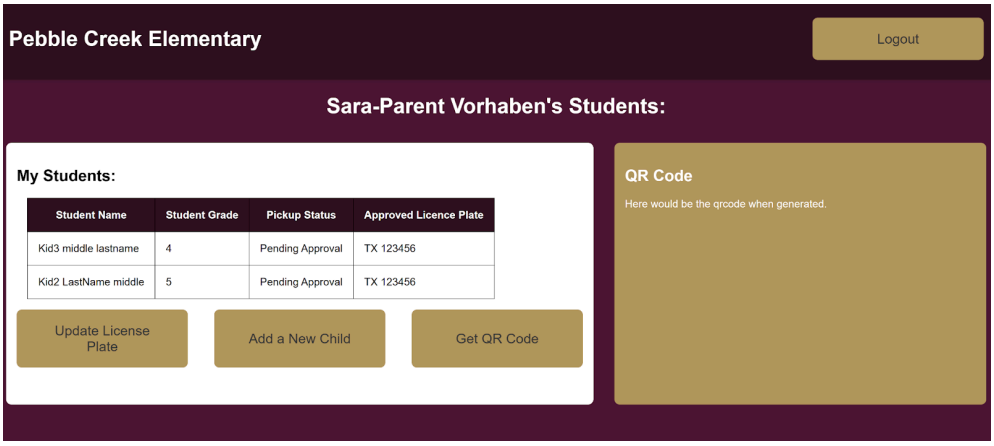
Teammates Present: all teammates present

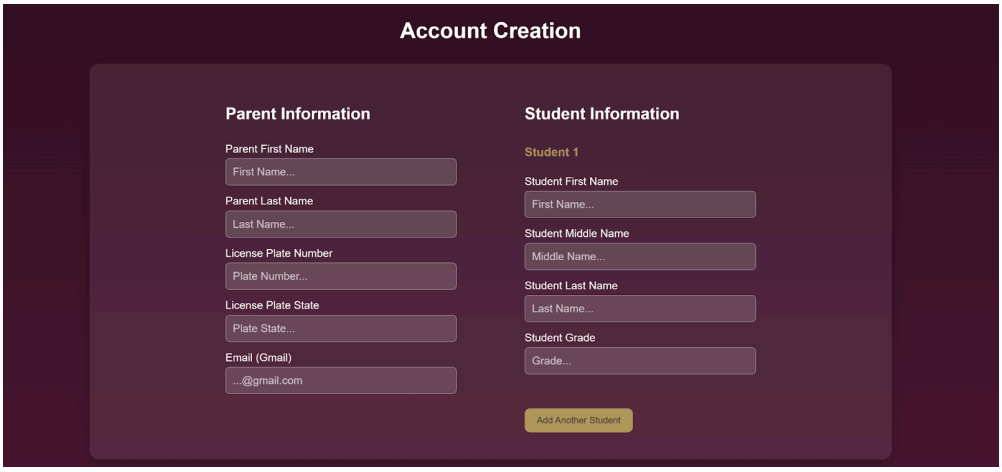
Updates on Previous Action Items:
1. Connect Database: completed
2. Complete Parent Portal page: done!
3. Look into QR APIs to implement: woking on later

What was accomplished in class:
In class, I worked on the css and connecting the tables to the database and using rls. Here

are some images showing my progress:







The next steps are to make the forms submit to the database for approval and for the pages to be more responsive. My goal is to work on that next week!
Jayden and Jack worked on the Zigbee connections and downloaded software to ensure that everything could be connected.
Additionally we worked on the weekly report so it could be submitted.

<u>Action items before next class:</u>

1. QR Research
2. Connecting other tables
3.

Questions for Teammates/Advisors:

Week 8:

Entry: #18

Class Journal Entry: March 02, 2026

Teammates Present: All teammates present

Updates on Previous Action Items:
1. QR Research: not much done
2. Connecting other tables: in progress

What was accomplished in class:
- Made it so the forms on the parents portal update the backend of the database
- Jack worked on licence plate reader
- Made it so table on parent portal dynamically updates

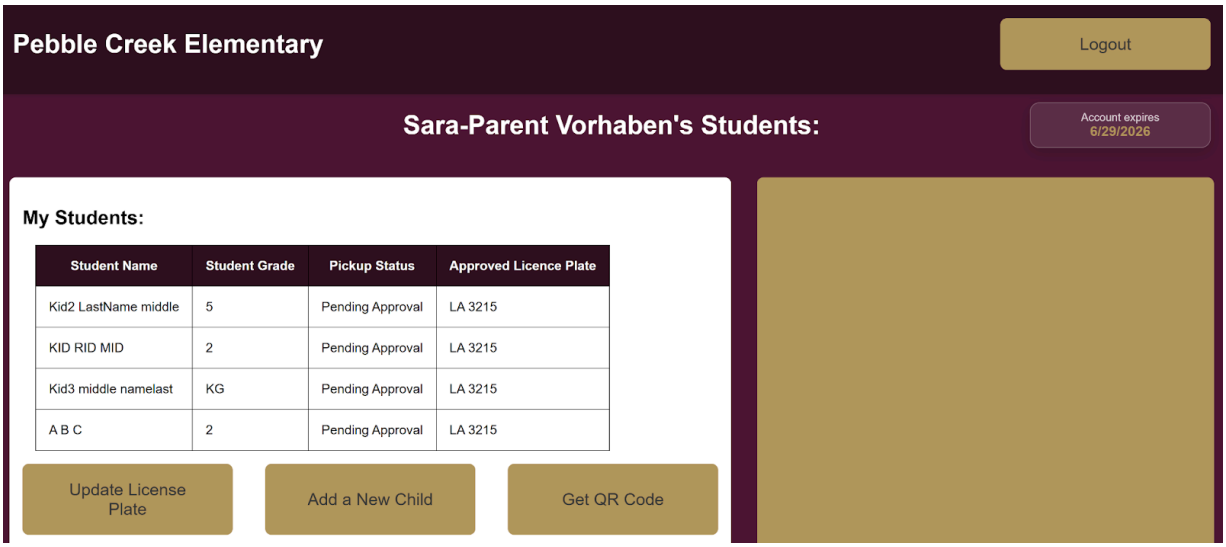
Entry: #19

Class Journal Entry: March 04, 2026

Teammates Present: All teammates present

What was accomplished in class:
- Website is almost done
- Worked on auth issue (protecting pages)
- Finished connecting things to the backend
- Began working on splitting up CDR presentation
- Began splitting up CDR report

Photos of progress on our website:



- Added expiration date
- Added loading when the site is getting the data so that it doesn't appear weird

<u>Action items before next class:</u>
1. Qr research
2. cdr
3.

<u>Project Status:</u>
Behind **On Time** Ahead

Week 9

Entry: #20

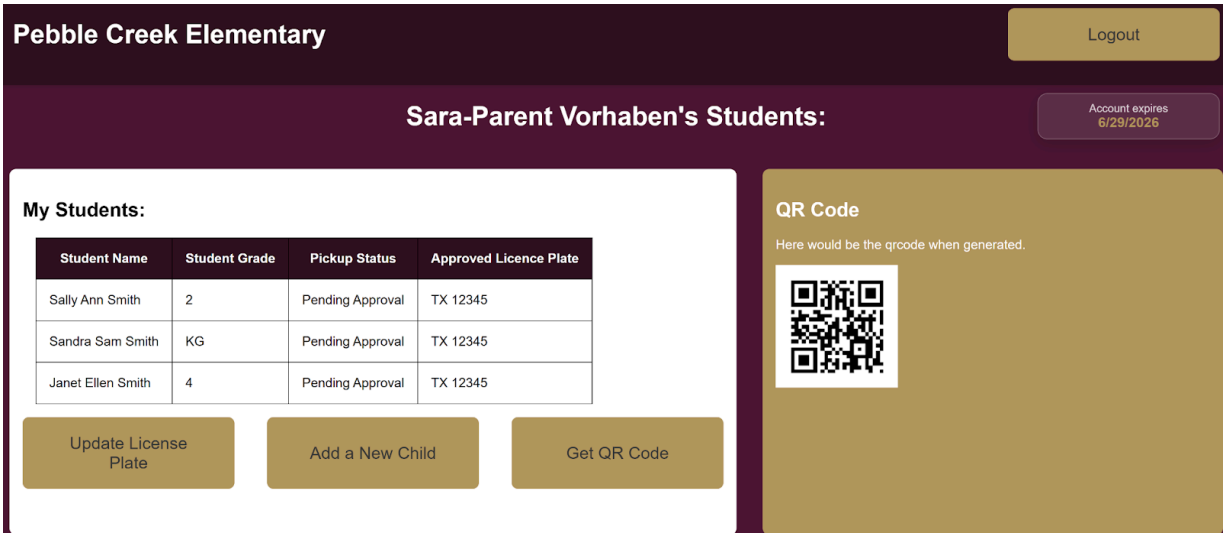
Class Journal Entry: March 16, 2026

<u>Teammates Present:</u> All Teammates Present

<u>Updates on Previous Action Items:</u>
1. Qr research: looked into qr codes
2. cdr: finished presentation and report

Project Status:
Behind **On Time** Ahead:

What was accomplished in class:
In class we presented our CDR presentation
Then I worked on finalizing the CDR report to ensure all requirements were included
I also looked into and began implementing QR codes.



Entry: #21

Class Journal Entry: March 18, 2026

Teammates Present: All teammates (worked remote)

Project Status::
Behind **On Time** Ahead:

What was accomplished in class:
I worked on qr code formatting and setup so that the qr code would match directly with what our system needed. I hope to be done with this relatively soon and get the deployed site updated

Action items before next class:
1. Finish QR codes
2. Update site
3. Begin looking into server integration

Week 10

Entry: #22

Class Journal Entry: March 23, 2026

Teammates Present: All teammates Present

What was accomplished in class:
Met with the Prof and talked about our progress.
We talked about website updates and showed him the QR code we added
Emailed School to meet and discuss our design

to mforester, Jayden, Jake, Jack, Haya, Swayamprava, Rabi

Howdy Coach Forester,
My name is Sara Vorhaben and I am a senior at Texas A&M University. For my capstone, my team is working on a school pickup system to try and make the school pickup process more efficient. Professor Mahapatra told us that you work in the car rider line regularly, so we feel you would be someone to reach out to regarding our system. My team and I wanted to reach out and get your opinion on some things and see if you had any suggestions or thoughts regarding our system to make it better and more user-friendly. Would you be free sometime this week to meet with our team via zoom or in person to talk for a little bit about our design and give us feedback? We would be free to meet Wednesday anytime between 11:30-2:30, but if there is another time that would work better, let us know!
Thanks,
Sara Vorhaben and The Wranglers

Action items before next class:
1. Look at Server setup and responsive design

Project Status:
Behind **On Time** Ahead

Entry: #23

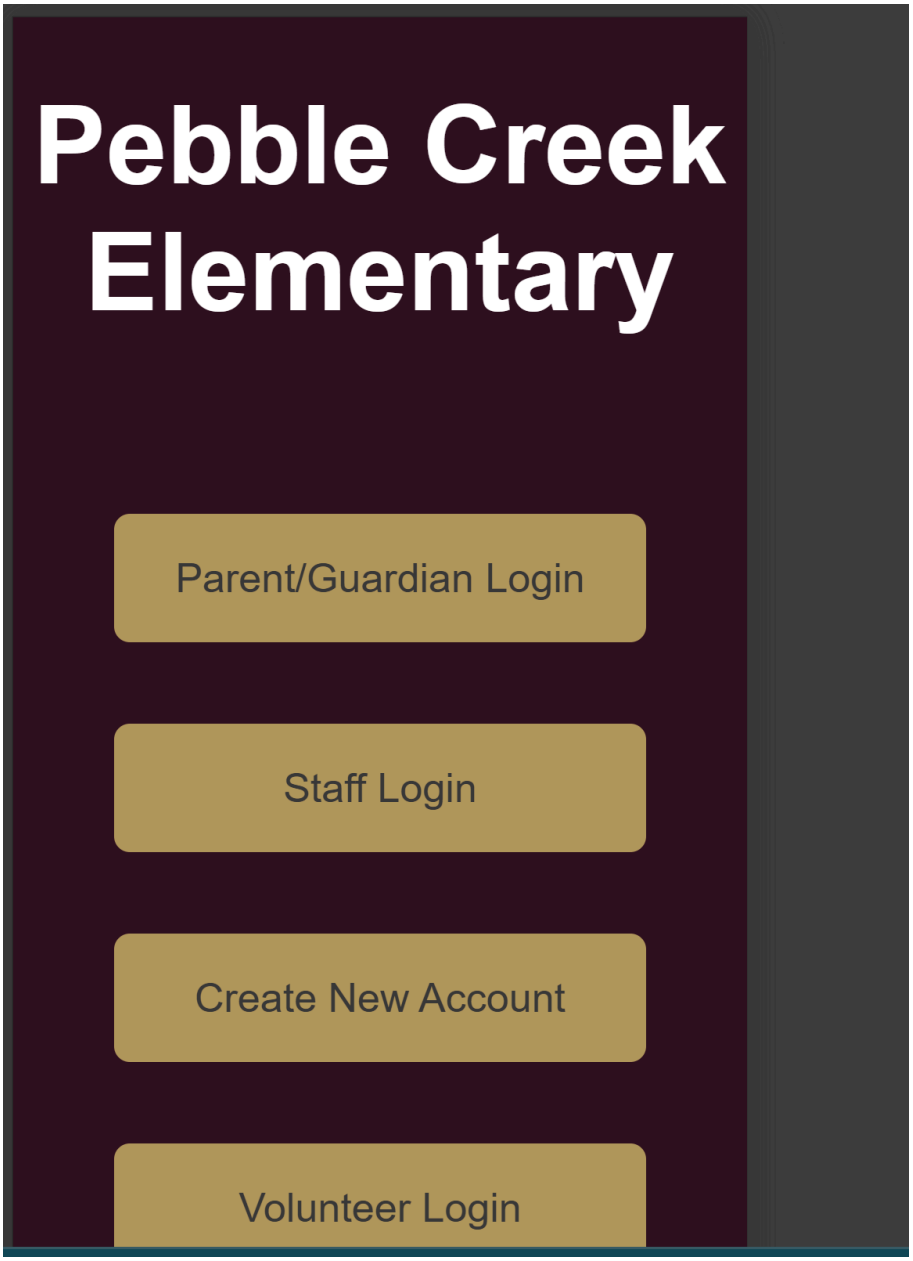
Class Journal Entry: March 25, 2026

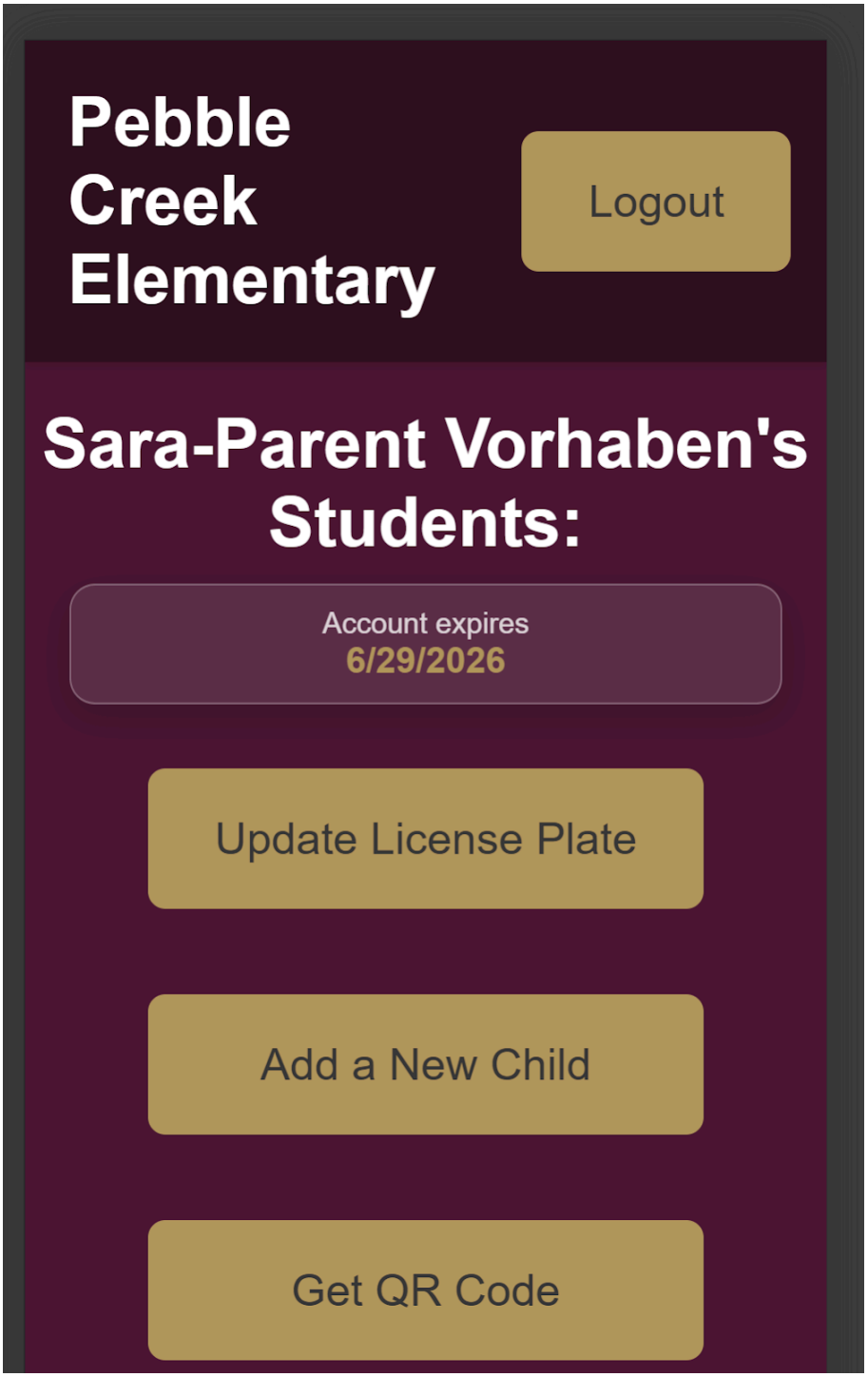
Teammates Present: Remote work day (all teammates present)

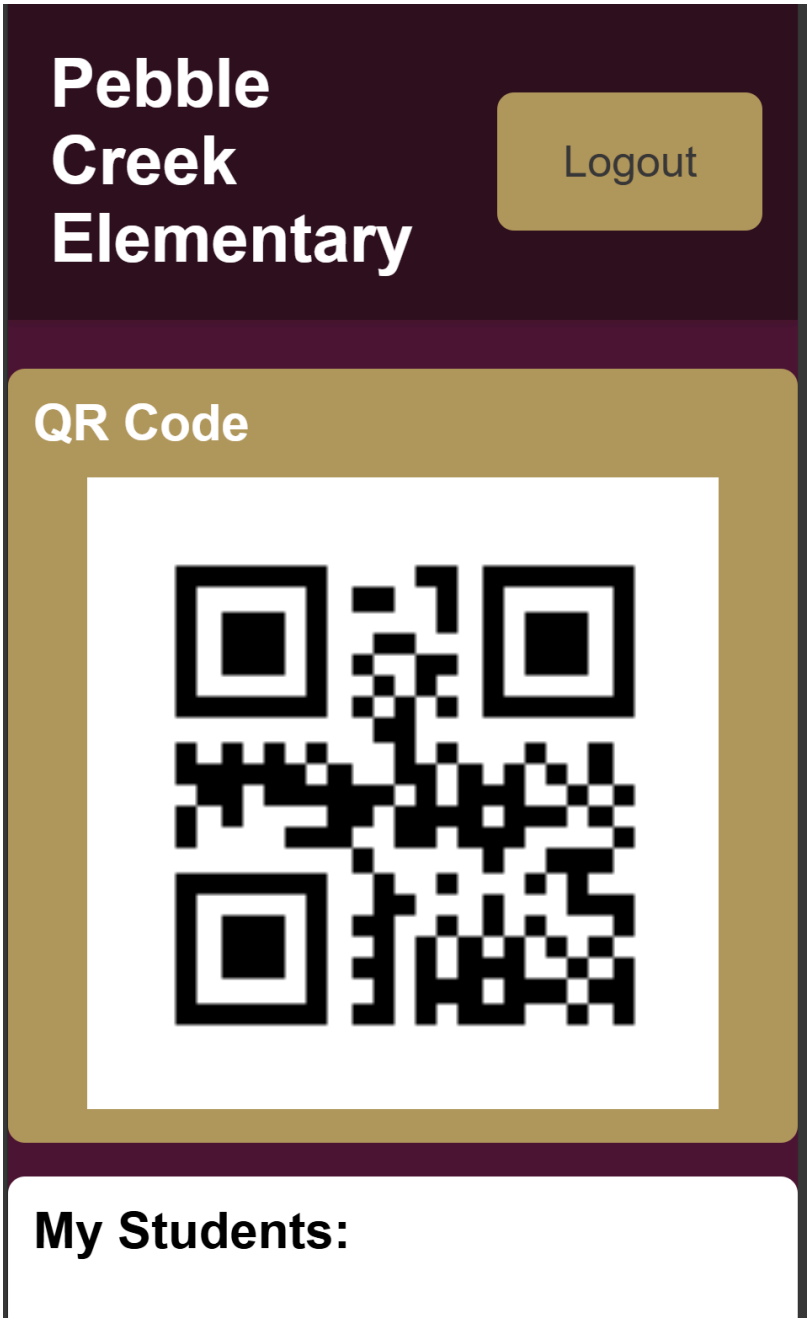
Updates on Previous Action Items:
1. Look at Server setup and responsive design: Watched youtube videos to try and figure it out

What was accomplished in class:
Set up the server and connected it to the website. Data can be sent from the server to the site and can be displayed on the site.
Updated the design of the parent portal and homescreen to make it responsive. The layout changed a little (especially on the parent portal) in order to make this work. This was done just

so the QR code would show up on the top and parents wouldn't have to unnecessarily scroll to see it.




Pebble Creek Elementary
Logout
Sara-Parent Vorhaben's Students:
Account expires
6/29/2026
Update License Plate
Add a New Child
Get QR Code



Action items before next class:

1. Push commits and comment code

Week 11

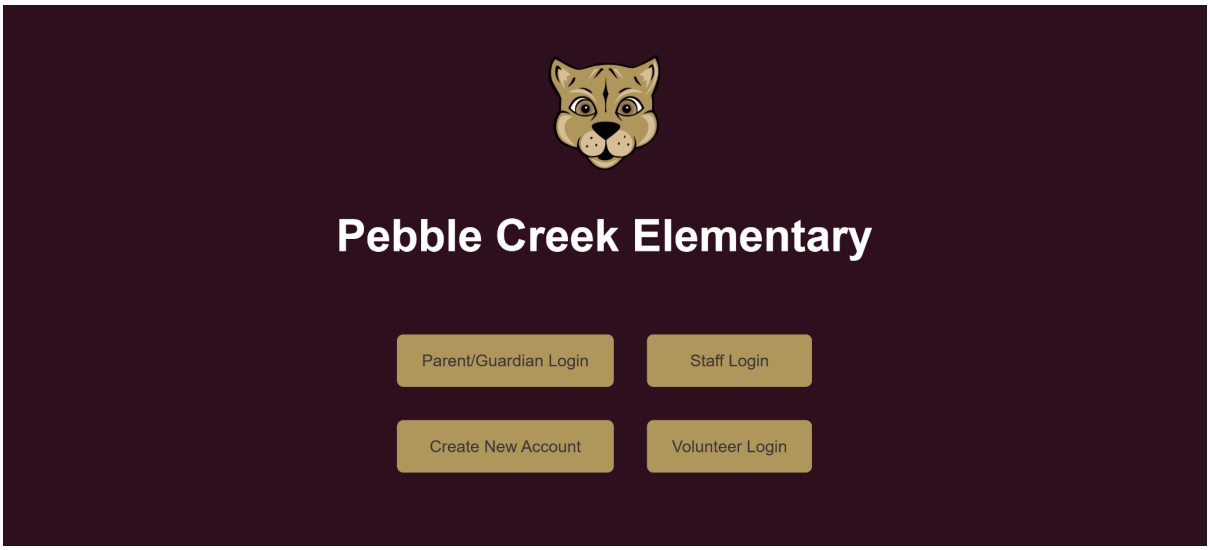
Entry: #24

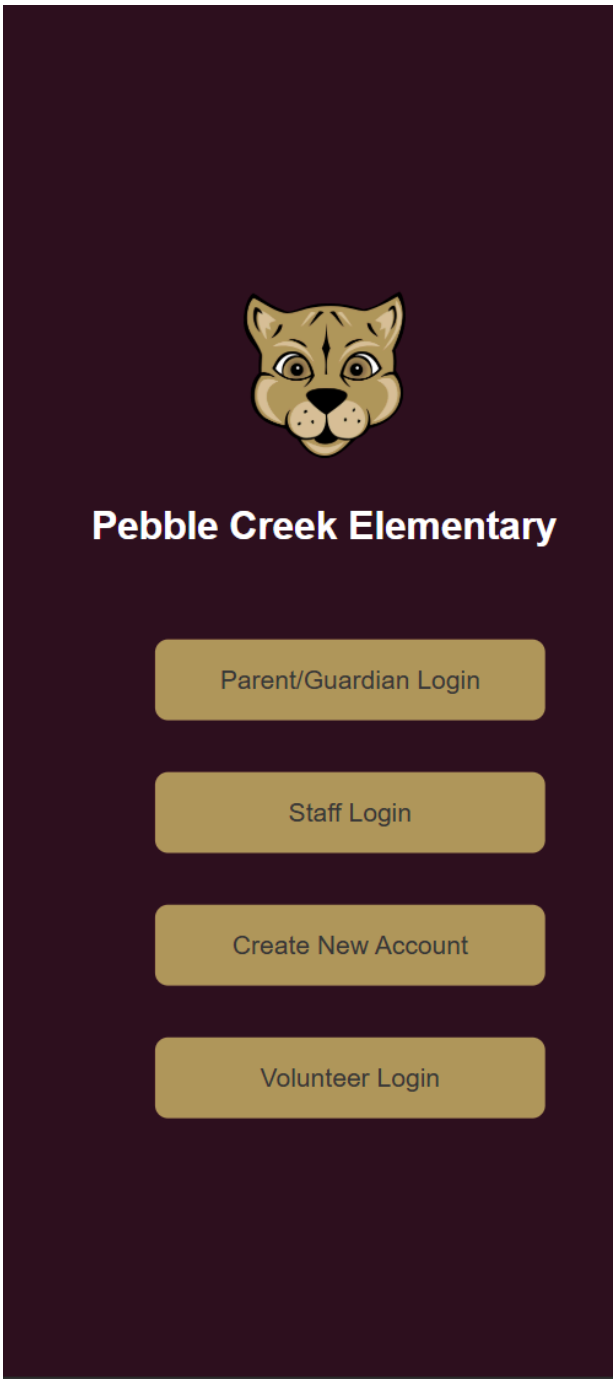
Class Journal Entry: March 30, 2026

Teammates Present: All Teammates Present

Updates on Previous Action Items:
1. Push commits and comment code: Committed

What was accomplished in class:
Met with prof:
- We talked about qr code scanning and demonstrated progress
- We showed our responsive design- I updated home and parent portal designs





- We talked about the 3d printed box and determined areas for improvement

In class I also worked on server setup to ensure that the server was working correctly.

Entry: #25

Class Journal Entry: April 01, 2026

Teammates Present: All Teammates present except Hiya (sick)

What was accomplished in class:
Had a meeting with the coach from the school on Zoom. All meeting notes can be found here
Action items before next class:
1. Continue server setup

Project Status:
Behind **On Time** Ahead

Week 12

Entry: #26

Class Journal Entry: April 06, 2026

Teammates Present: All teammates present

Updates on Previous Action Items:
1. Continue server setup: continued working on server setup and determined I had to use NAT instead of bridged bc the school wifi wont allow the additional connection on Bridged devices (so it shows up as no wifi/ethernet and wont let you do anything), I had to reset the server in order to fix this and we are going to have to do NAT with port forwarding instead

What was accomplished in class:

- Set up NAT with port forwarding and was able to successfully ping a device
- Met with the professor and discussed our project progress
    - 3d printing assembly changes
    - Qr code setup
    - Updates to structure of server
    - Meeting with school
- Worked on connections to ensure that everything will work together, was able to get the server to register input and output from the website locally, going to push and see if everyone else can access and get it to work on their devices
- Submitted housing for 3D printing

Action items before next class:
1. Figure out connection issues and create updated website to fix it

Project Status:
Behind **On Time** Ahead

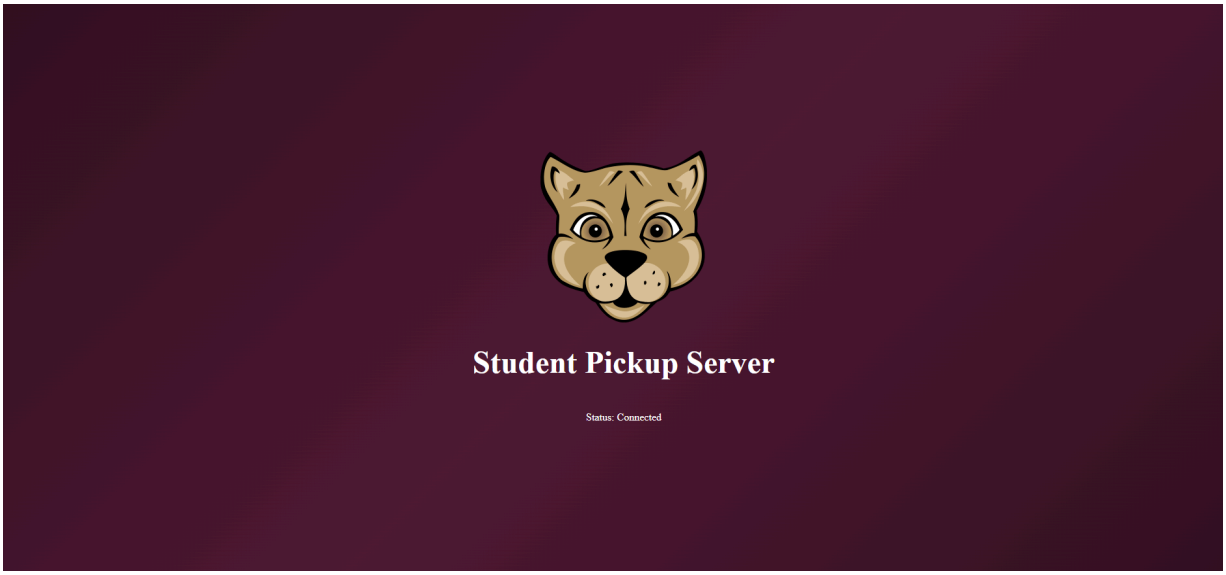
Entry: #27

Class Journal Entry: April 08, 2026

Teammates Present: All teammates present except Jack

Updates on Previous Action Items:
1. Figure out connection issues and create updated website to fix it: fixed

What was accomplished in class:
Set up a server so that it now works across devices. It was deployed on render here: https://wranglers-capstone.onrender.com/



I created endpoints so that the server can connect to the website and display data
Additionally, set up and endpoint for testing so that the website can send data to the server if the pi is not present.

Week 13

Entry: #28

Class Journal Entry: April 13, 2026

Teammates Present: All Teammates present

What was accomplished in class:

- Met with prof - talked about 3D print needing to be redone, server setup updates
- Worked on integration- was able to get a response from Zigbee?

Action items before next class:

1. Work on formatting/ handling requests with new format and connection
2. Begin Testing

Project Status:

Behind **On Time** Ahead

Entry: #29

Class Journal Entry: April 15, 2026

Teammates Present: All teammates present

Updates on Previous Action Items:

1. Work on formatting/ handling requests with new format and connection: working!!
2. Begin Testing: working!!

What was accomplished in class:

Full system test:sent QR Code to the website and displayed results.

- It worked (sort of) the qr code wasn't scanned live, just the data was passed because these weren't connected yet, but the server was able to handle successful cases and display them correctly
    - More error management needs to be done to account for failures to make it more user friendly

Action items before next class:

1. Make it pretty

Project Status:
Behind On **Time** Ahead

Questions for Teammates/Advisors:

Week 14

Entry: #30

Class Journal Entry: April 20, 2026

Teammates Present: All Teammates Present

Updates on Previous Action Items:
1. Make it pretty: Site and formatting look nice

What was accomplished in class:

- 3D print arrived! Looks nice and works on ensuring all components fit nicely and will work with the tripod.

- Full System test: Scanned LP, updated server, updated site, cleared kid
- Fixes to account for QR not in the database, unapproved students.
- Fixes to display screen on the PI

Action items before next class:
1. Acquire a USB A to USB C cord
2. Acquire a USB

Project Status:
Behind **On Time** Ahead

The project is mostly done. Only Testing Remains

Entry: #31

Class Journal Entry: April 22, 2026

Teammates Present: All Teammates Present

Updates on Previous Action Items:
1. Acquire a USB A to USB C cord:
2. Acquire a USB:

What was accomplished in class:
- Met with the prof and demoed our product - it worked!
- Got feedback to add sound to pickup to announce who was being picked up inside student pickup module
- Wrote code to add sound to module, increase the length of time that the screen station was shown
- Met up as a team and tested whole system with cars (both a truck and a SUV)

Action items before next class:
1. Complete Section of the Report
2. Complete Presentation Slides
3. Compile the deliverables

Project Status:
Behind **On Time** Ahead

Entry: #32

Personal Work Journal Entry: April 27, 2026

Teammates Present (if any): Only Me

Updates on Previous Action Items:
1. Complete Section of the Report: I worked on the server and database implementation as well as all of the results and analysis section
2. Complete Presentation Slides: I created the final presentation slides and edited my portion of the slides
3. Compile the deliverables: I created a final deliverables folder in our google drive and added all of our reports to it.

Later in class we will present our slides :)